Introductory activity

Activity 1 · One root hides two others 35 min . Consider $P(z) = z^3 - 2z^2 + 4z - 8$.

1. Plot on a calculator the curve of $x \mapsto x^3 - 2x^2 + 4x - 8$ on $[-3, 4]$. How many real solutions does the equation $P(x) = 0$ seem to have? Locate one.
2. Check that $P(2) = 0$.
3. Find real numbers a, b, c such that, for every z, $P(z) = (z - 2)\left(az^2 + bz + c\right)$. Expand the right-hand side and identify the coefficients.
4. Deduce all the complex solutions of $P(z) = 0$.
5. Compute the sum and the product of the three solutions. Compare with the coefficients of P.
6. How many complex solutions can a polynomial of degree 3 have at most? State a conjecture for degree n.

Observing is not proving: the properties conjectured here are established in the course.